

FlowSync: Intelligent Real-Time Streaming Analytics

An End-to-End Data Engineering & AI Platform for Streaming Events

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Fundamentals of Data Engineering Project Review-2

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Abstract

FlowSync is an end-to-end real-time Data Engineering platform designed to handle continuously generated, high-volume streaming data with minimal latency. It enables organizations to build a reliable pipeline for transforming raw streaming events into clean, structured, and AI-ready data. The platform ingests real-time data from diverse sources such as IoT devices, application logs, and transactional systems using Apache Kafka. Spark Structured Streaming processes these events in real time by performing data validation, cleansing, transformation, and enrichment.

Processed data is securely stored across MinIO S3 Lake, Parquet files, and PostgreSQL, providing scalable storage as well as easy access for downstream analytics and applications. FlowSync also incorporates AI-based anomaly detection to identify unusual patterns, potential failures, and abnormal events as they occur.

With the rapid expansion of IoT ecosystems, connected devices, and real-time digital services, organizations are generating massive amounts of data every second. Traditional batch-processing systems often introduce significant delays, making it difficult to detect critical events and take timely action. FlowSync addresses this challenge by providing a low-latency, scalable, and fault-tolerant streaming architecture that continuously converts raw data into actionable intelligence. By combining real-time data engineering with AI-driven analytics, FlowSync helps organizations achieve faster insights, proactive anomaly detection, and improved decision-making.

Modern applications generate large volumes of continuous streaming data from APIs, applications, IoT devices, and operational systems. However, traditional batch-processing architectures introduce high processing latency, making it difficult to obtain timely insights for real-time monitoring and anomaly detection. At the same time, streaming data often contains missing values, duplicate records, schema inconsistencies, and malformed inputs, which can reduce the reliability of downstream analytics and AI models.

Furthermore, data is frequently distributed across separate storage, processing, and analytics environments, creating silos and making real-time data management difficult. There is a need for an integrated solution that can ingest streaming data in real time, validate and clean the data, detect anomalies, store the processed data locally, and provide unified visualization and monitoring.

Therefore, the proposed system aims to develop a real-time streaming data quality and anomaly detection framework that combines stream processing, automated data validation, anomaly detection, local data storage, and dashboard-based analytics to provide faster, cleaner, and more reliable insights from continuously generated data.

Research Objectives

- **Low-Latency Ingestion** – Build a scalable Apache Kafka pipeline to capture high-volume IoT and application events.
- **Real-Time Stream Processing** – Use Apache Spark Structured Streaming for continuous data transformations and aggregations.
- **Automated Data Quality** – Perform real-time schema validation, deduplication, and data cleaning using PySpark and Great Expectations.
- **Hybrid Analytical Storage** – Store raw and processed datasets using MinIO S3-compatible storage, Parquet, and PostgreSQL.
- **Intelligent Anomaly Detection** – Apply Scikit-Learn models to identify streaming outliers and unusual behavioral patterns automatically.
- **Live Serving & Monitoring** – Provide analytical REST APIs using FastAPI and real-time visualization and monitoring through Grafana.

Literature Survey



Authors & Year	Paper Title	Methodology	Key Inferences	Limitations / Gaps
K. Zhang et al. (2022)	Real-Time Streaming Analytics Architecture	Apache Kafka + Spark Streaming for log ingestion	High throughput stream collection and transformation	Lacks inline streaming data quality validation frameworks
M. Gupta & R. Kumar (2023)	Data Quality Assurance in High-Velocity Streams	PySpark with schema enforcement mechanisms	Identifies dirty streaming data before warehouse ingestion	High computational overhead on complex ML feature pipelines
S. Patel et al. (2024)	Edge-Driven Stream Anomaly Detection	Scikit-learn Isolation Forest on continuous events	Effective detection of real-time operational outliers	Omitted local S3 storage and unified Grafana dashboard integration
A. Sharma et al. (2024)	Real-Time Data Quality Monitoring in Streaming Pipelines	Apache Kafka + PySpark with automated validation and anomaly detection	Enables continuous monitoring of streaming data quality and detects invalid, missing, and anomalous records in real time	Limited integration with local S3 storage and lacks a unified visualization and alerting dashboard

Technology Stack

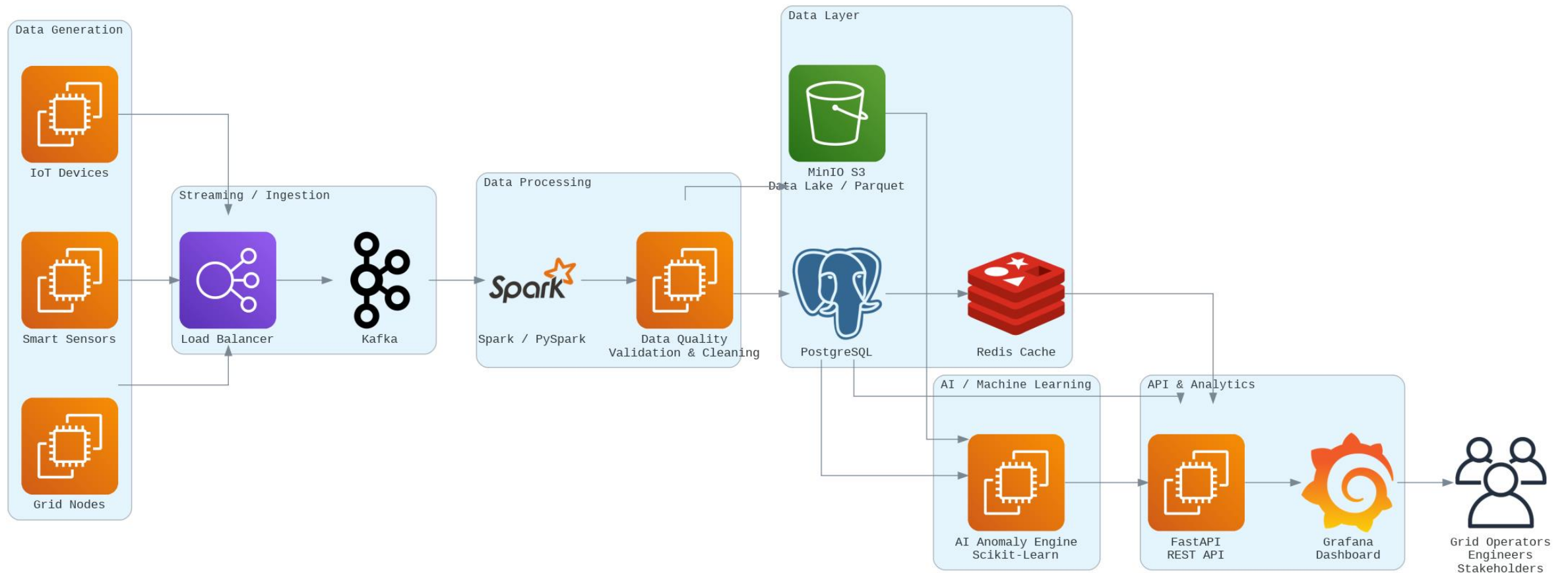


Category	Technology	Purpose
Programming	Python, PySpark	Pipeline development and distributed data processing
Data Ingestion	Apache Kafka	Real-time ingestion and event streaming
Stream Processing	Spark Structured Streaming	Low-latency processing and transformation of streaming data
Data Quality	Great Expectations	Data validation, profiling, and quality checks
Data Lake	MinIO (S3-Compatible)	Scalable object storage for raw and processed data
Data Storage	Apache Parquet, PostgreSQL	Efficient analytical storage and structured data persistence
Machine Learning	Scikit-learn	Anomaly detection and machine learning-based analysis
API Layer	FastAPI	High-performance APIs for data and ML services
Monitoring	Grafana	Pipeline, system, and real-time performance monitoring
DevOps & Version Control	Docker, Git	Containerization, deployment, and source-code management

Proposed Solution

- **Real-Time Data Ingestion & Processing:** Apache Kafka collects real-time data from multiple sources and streams it into the pipeline. PySpark and Spark Structured Streaming process and transform the incoming data efficiently. The system supports low-latency processing for continuous data analysis. This ensures reliable and timely data availability.
- **Data Quality & Storage:** Great Expectations validates the incoming data and performs schema, consistency, and quality checks. MinIO provides scalable S3-compatible storage for raw and processed datasets. Apache Parquet enables efficient storage and analytical processing. PostgreSQL is used for structured and relational data persistence.
- **AI, API & Monitoring:** Scikit-learn models analyze processed data to identify anomalies and unusual patterns. FastAPI provides high-performance APIs for accessing data and machine learning services. Grafana dashboards display real-time system performance and detected anomalies. This enables effective monitoring and faster decision-making.

End-to-End System Workflow



Smart Grid - Data Architecture

Thank You!

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